

LAST NAME: _____

Practice Quiz 02A — MATH 2055 — Fall 2026

Practice for Quiz 02

The following is closed book and closed note. Don't assume the grader will know what you mean. Use full sentences and provide complete and correct solutions.

Revised using GPT-6 Astra in the Codex harness.

1. (a) Write the converse and contrapositive of $\neg P \rightarrow Q$. Make one complete four-row truth table comparing all three formulas, and state which of the two new formulas is equivalent to the original.

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- (b) In $\exists t \in \mathbf{R} (t^2 + c = w)$, identify the free and bound variables.